

ZACHARY ONGERI

AI NATIVE SOFTWARE DEVELOPER // AGENTIC & LLM SYSTEMS // ECOMMERCE // WEB INFRASTRUCTURE

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WHAT I BUILD

I take products from an empty repository to something a business runs on. Retrieval systems and agents, storefronts I design as well as build, and the infrastructure underneath both. Founder operator rather than agency, so I have carried the cost of my own bad architecture decisions. **Every line of code should move a balance sheet number.** If it does not cut cost or raise transaction value, it is clutter.

SELECTED BUILDS

01 AGENTIC & LLM SYSTEMS

RegWatch

Multi tenant regulatory intelligence platform, built inside MNL Advocates LLP · [dates](#)

- Full retrieval pipeline end to end: Voyage AI embeddings, Supabase pgvector store, Next.js 14 application layer, covering Central Bank of Kenya and Data Protection Commissioner filings.
- Three role row level security model so one tenant can never read another's filings. Isolation sits in Postgres policies rather than application logic, which is the difference between a guarantee and a good intention.
- Owned the whole surface: ingestion, chunking, embedding, ranking, auth, front end, deploy.

Vertical AI Agents for Clinics, Dealerships and Short Stay Hosts

Production deployments across Kenyan and United States markets · [dates](#)

- Conversational agents on the WhatsApp Business API and Claude API that answer inbound enquiries and qualify leads in under 30 seconds, with handoff to a human when the agent is out of its depth.
- Built narrow and vertical rather than general purpose. An agent that only handles booking, triage or stock questions can be tested, corrected and trusted.

02 ECOMMERCE, DESIGN THROUGH BUILD

FORMA

Direct to consumer apparel brand, founded and built from the sourcing layer up · [dates](#)

- Owned the full stack of the business rather than just the site: supplier sourcing locked to S-Shaper at OEKO-TEX certification and a 100 unit MOQ, sizing rebuilt around East African proportions, plus a full identity system and Shopify storefront built before the first unit shipped.

Naisole

Shopify storefront, design and build · [dates](#)

- Store designed and developed end to end: [category, theme or customisation detail, launch outcome](#). Same pattern as FORMA, with identity, information architecture, product data model and conversion path treated as one system rather than four handoffs.

03 WEB INFRASTRUCTURE & SEARCH

MNL Advocates LLP Platform Migration

Headless rebuild and search rearchitecture · [dates](#)

- Migrated the firm's web presence to a headless stack: WordPress as CMS, Next.js front end, Vercel edge delivery. Architecture, build, redirect mapping, cutover and post launch QA were all mine.
- Rebuilt SEO and AEO from the ground up rather than porting legacy debt, structuring the firm and its practice areas as entities so the site reads cleanly to answer engines as well as crawlers. Result: [traffic or Search Console result](#).

04 PRODUCTIVITY & OPERATING SYSTEMS

Internal tooling built alongside the client work it supports: lead capture and routing, scheduled sourcing and lead generation sweeps, and event operations for the Africa Leadership Circle across the Nairobi summit and the Benin breakfast in Cotonou.
The pattern: most small business software problems are not missing features, they are a manual step that runs weekly and nobody has costed.

HOW I WORK

One person, whole surface. I do not hand a design to a developer or a spec to an agency. Identity, data model, application and deployment get decided together, which is why the seams do not show. If a project needs a team, I will say so instead of stretching.

STACK

AI & RETRIEVAL

Claude API | Voyage AI | pgvector
RAG pipelines | Agent design | Evals

APPLICATION

Next.js 14 | React | TypeScript | Node
Tailwind

DATA & BACKEND

Supabase | PostgreSQL | Row level security
SQL

COMMERCE

Shopify | Storefront design
Product data modelling | Supplier sourcing

INFRASTRUCTURE & SEARCH

Vercel | Headless WordPress
Core Web Vitals | Schema.org | GA4 / GTM
Search Console

INTEGRATIONS

WhatsApp Business API | Webhooks
Scheduled jobs

DOMAINS SHIPPED IN

Legal and regulatory compliance, apparel and direct to consumer retail, healthcare clinics, vehicle dealerships, short stay hospitality, professional services.

EDUCATION

University of Nairobi

AVAILABILITY

Remote from Nairobi, UTC+3. Overlap held open for European, United States and Asia Pacific working hours.

BUILD LOG

ZACHARY ONGERI // DECISIONS, METHOD, ENGAGEMENT SHAPES

The part of a portfolio that usually goes unwritten. Read this if you want to know how I think before you hire me.

DECISIONS I HAVE MADE AND WHY

Tenant isolation belongs in the database, not the application.

RegWatch holds filings for competing clients. I put separation in Postgres row level security with three roles rather than filtering in application code. Application filters are one forgotten **where** clause away from a breach. A policy at the data layer fails closed.

Narrow agents beat general assistants in production.

Every agent I have shipped does one job for one vertical. A booking agent for hosts, a triage agent for clinics, a stock and finance agent for dealerships. Narrow scope means you can define what correct looks like, test against it, and hand a business something it can trust with its own customers.

Headless was a search decision before it was a speed decision.

Moving MNL to WordPress plus Next.js on Vercel was not about the stack being current. Editors keep an interface they already know, while the front end gains control over rendering, structured data and edge delivery. That control is what makes a site legible to answer engines rather than merely fast.

In ecommerce, the supply chain is part of the software.

With FORMA I locked the supplier, the certification and the MOQ before designing a single screen, and rebuilt sizing around the customer rather than an inherited size chart. A storefront built on top of a broken product decision converts badly no matter how good the build is.

HOW I BUILD AN AGENT

1. Find the conversation that already happens and read a hundred real messages before writing anything.
2. Define the narrow job and the exact point where the agent must hand off to a person.
3. Build retrieval before personality. Wrong facts in a friendly voice are worse than no agent.
4. Write the failure cases as tests, then run them every time a prompt changes.
5. Ship to one channel, measure the handoff rate, and only widen scope once it falls.

ENGAGEMENT SHAPES

Build sprint. A defined product or storefront taken from nothing to live, fixed scope and milestones.

System audit. A read of an existing stack, its search surface and its manual steps, delivered as a prioritised plan you can execute without me.

Retained build. Ongoing ownership of a product's technical direction for teams without an in house engineer.

HOW A BUILD SPRINT RUNS

WEEK 01 / DEFINE

The commercial outcome stated as a number, not a feature list. Data model, integration surface and the one metric the build has to move, agreed in writing before any code exists.

WEEK 02 - 05 / BUILD

Working software in front of you every week rather than a reveal at the end. Real data early, because demo data hides every problem worth finding.

WEEK 06 / HAND OVER

Deployment, tracking wired to real events, and written documentation your team can run. If you need me afterwards it should be for the next build, not to keep this one alive.

I run RegWatch inside a law firm, build FORMA from the sourcing layer up, and trade my own capital. I build for clients the way I build for myself.